Position: earn the right to evaluate

Decision record · synthetic public edition · 7 November 2026 (Europe/Berlin)

1. The product is the evidence chain

A cold-chain alarm is not a product disposition. A useful system makes a reviewer faster at establishing what can legitimately be concluded, rather than generating a confident answer from incomplete inputs. The sequence is candidate discovery, exact identity/strength/presentation applicability, source-content verification, governing authority, policy evaluation, then an authorized human disposition. These are separate gates, not synonyms.

A manual can be readable but inapplicable. A matching product name can identify the wrong presentation. An index can locate a document without establishing its authority. Two effective SOP revisions cannot be resolved simply by selecting the largest revision number. The software must expose these distinctions.

The public implementation chooses a difficult but narrow slice: a tenant-bound evidence workbench with a durable, independent notification ledger. It captures attached source bytes, local simulated approvals, explicit arithmetic, unresolved tasks and reproducible snapshots. It deliberately has no real clinical disposition endpoint. This prevents a compelling demo from pretending to be a qualified release system.

2. What is actually built

FastAPI provides a typed HTTP boundary; Pydantic rejects unexpected fields and invalid measurement coercions; a pure Python evaluator performs explicit synthetic comparisons; SQLite makes transactions and restart behavior easy to inspect. The UI is plain HTML/CSS/JavaScript. The runtime needs no external service. A reviewer can start it, inspect code and exercise negative cases without learning an agent stack.

An invented teaching policy has explicit lower/upper operators, an individual event duration ceiling and a separate cumulative ceiling. It explicitly asserts the relationship for that fictional demonstration. This is not an inference about any real product. In real evidence, individual and cumulative constraints may coexist; neither a phrase describing one excursion nor a lifetime allowance specifies their full interaction. Unknown history is never zero. Unmodeled constraints block supported evaluation.

The three fictional cases demonstrate usable supported arithmetic, unresolved authority, and an out-of-envelope result. All preserve source references and inputs. A missing source, blank transcription or unapproved record creates a role-assigned resolution task. That is useful work, not merely a BLOCKED flag.

3. Boundaries that matter more than framework count

Two clocks. Evidence review and notification are independent. Every imported case creates an UNKNOWN obligation even if its raw flag says N. Only the Regulatory role can record a qualified assessment. A QA acknowledgement never updates the obligation ledger. A failed evidence assembly does not prevent the queue from loading. Deadline ordering uses explicit offset-aware deadlines; no invented legal calendars. The demonstration records notification assessments, but sends nothing.

Two kinds of trust. Authentication establishes a locally provisioned actor; the database binds that actor to tenant and role. A caller does not acquire membership by passing a tenant string. Local role keys are generated per runtime, expire, and are stored as hashes server-side. Human review labels are not identity proof. This is not an OIDC identity system or a qualified e-signature implementation.

Time and versions. Immutable case inputs are preserved. New attached sources and approvals create a new source revision, and a new evaluation creates a linked bundle. Old results are not overwritten. Replay uses captured inputs and the supported evaluator version, never later outcomes or current source records. Reassessment uses current evidence and remains distinguishable from original replay.

Races. Acknowledgement verifies the stored bundle hash, client expected hash, latest version, current source revision and current source bytes in one transaction. One acknowledgement per bundle and a scoped idempotency constraint handle retries and racing reviewers. An unapproved newly attached source also makes an old snapshot stale: evidence arrival is significant even before authority is established.

Integrity is not truth. Hash checks detect content mismatch against stored expectations. An attacker with full database write control can rewrite both data and hashes. Local append-only triggers are useful application defenses, not WORM storage, cryptographic signatures or attestation of a study's truth.

4. The intentional non-goals

No medicine release, destroy, quarantine or shortened-expiry computation. No autonomous alert closure. No live source connectors, OCR, fuzzy identifier merging, raw-series completeness inference, legal notification-window engine, regulatory message delivery, commercial claims or shared precedent index. No multilingual regulated output, dual signatures, MFA, validated retention or multi-region deployment. These belong in a qualified broader system, not half-built around a deceptively impressive demo.

A future retrieval assistant may propose sources; its output remains a candidate and must pass the same gates. It cannot choose governing authority or invent missing evidence. No LLM is needed for the selected slice, which removes cost, leakage and model-change dependencies from the trusted calculation path.

5. Delivery and production qualification

Start by pinning unresolved factual questions and original acceptance evidence privately. Deliver the position and system design, then implement only the chosen difficult slice with negative tests and truthful execution records. Use a focused synthetic walkthrough, not hundreds of shallow framework files.

For a larger team, split ownership into source intake/identity, policy authority, workbench/evaluation, notification operations, platform/security, and independent QA/replay qualification. Contracts in the LLD are the shared boundary. Each component must fail without silently changing the meaning of another.

A production pilot needs explicit approved identity mappings, authoritative SOP/annexure selection, complete history and telemetry, customer-specific professional authority and reporting calendars. Before each change, an independent QA-reviewed corpus must compare old and new behavior with explained differences and denominators. Historical labels are evidence about past decisions, not an oracle. Unreviewed closures and non-random samples cannot establish production performance.

Infrastructure follows demonstrated bottlenecks. SQLite intentionally serializes this local slice. The design moves to regional PostgreSQL with tested tenant row policies, separately scoped object storage, durable outbox delivery and fair per-tenant scheduling before multi-host deployment. That migration requires new concurrency, recovery and isolation tests; changing a connection string is not sufficient.

6. Measure value without pretending it exists already

Separate targets from achievements. Measure signed-disposition lead time only once real professional signing exists; notification deadline compliance separately; evidence retrieval completeness and time; reason-specific resolution time; override explanations; tenant access failures; and all-in running cost. A synthetic supported-path count is not an alert-closure rate. Local tests passing are not a safety case.

Budget model inference and its serving retrieval separately from ingestion, storage, indexing and retention. Record whether reviewer-stage allowances include screening and how shared retrieval is allocated. Price idle capacity as well as busy workloads. This demo has no model bill but still consumes CPU, disk and human review time; zero model cost is not zero operating cost.

Production go-live requires security review, qualified professional identity/signing, restore drills, regional residency controls, change qualification, and customer-specific acceptance. Those are gates, not promises earned by an architecture diagram or a Dockerfile.

Open questions: Which approved sources apply at the event's sites and time? Who resolves revision and annexure conflicts? How do repeated and cumulative exposures interact? Who attests that batch and telemetry histories are complete? Which programmes and calendars govern notifications? Which roles, delegations, language and signing factors apply? How are costs allocated? The public fixtures answer these only for their explicitly invented scenario.